

Lab 1 – Ideal Gas Law

CHE 260 • Thermodynamics and Heat Transfer

There Are 2 Parts To Complete In This Lab.

Before Beginning this Lab, be familiar with the following terms:

- **Absolute pressure** is zero-referenced against a perfect vacuum, so it is equal to gauge pressure plus atmospheric pressure.
- **Gauge pressure** is zero-referenced against ambient air pressure, so it is equal to absolute pressure minus atmospheric pressure. Negative signs are usually omitted.

**** When answering the discussion questions, remember to use the *absolute pressure* ****

Safety Warnings for this Lab:

- Use the flow meter control in LabVIEW to ramp up the flow of air GRADUALLY from zero. For example, DO NOT set the flow rate to maximum and open all the valves to fill a tank or to create a vacuum, instead, ensuring that the flow rate is set to zero initially, open all the desired valves and then ramp up the flow rate in increments to the maximum flow. This procedure ensures that the flow controllers do not get a large surge of flow through them
- Do not allow pressure to exceed 80 psi (always keep $P < 80$ psi)
- Do not allow temperature to exceed 70°C (always keep $T < 70^{\circ}\text{C}$)
- Always wear safety glasses when operating this system

Nomenclature for this Lab:

T_1 :	Left Tank Temperature [$^{\circ}\text{C}$]
T_2 :	Right Tank Temperature [$^{\circ}\text{C}$]
P_1 :	Left Tank Gage Pressure [Psig]
P_2 :	Right Tank Gage Pressure [Psig]
[Psig]	Gage Pressure [Psi], Pressure reading that you actually see on the gages
[Psia]	Absolute Pressure [Psi], $P_{sia} = P_{sig} + P_{atm}$

Part 1: Path Independency of Thermodynamics State Properties and Determining the Volume Ratio of the Tanks

Introduction

The idea in this lab is to expand air from a pressurized container into another container at different rates. In the first case, the pressures between the two tanks are equalized rapidly and time is given for the tanks to come to thermal equilibrium. In the second case, using the same initial conditions as in case one, the tanks are brought to equilibrium at a rate in which the tanks remain in thermal equilibrium throughout the equalizing process. The purpose is to show that even though the expansion process is different between the two cases, the final thermodynamics state properties, i.e. temperature and pressure, are the same. Hence, the state properties are path independent.

To begin a lab:

1. Turn the computer on and start LabVIEW (or click 'Reset' button if LabVIEW is already running)
2. Make sure all valves are closed.

*Note: Recall from the introductory lab that the **regulator pressure** will be preset to 80 psi.

Procedure

1. Record the ambient pressure using the manometer in the lab.
2. Click the 'Start Collecting Data' button in LabVIEW to begin recording the results on the graphs.
3. Pressurize the left tank to **40 psig**. The procedure to do this is as follows:
 - a. Open Valve A2.
 - b. Open the Left Solenoid valve by pressing the 'Left Solenoid' button in LabVIEW once.
 - c. Using LabVIEW, gradually increase the Flow Rate to 50 g/min.
 - d. When the pressure reaches **40 psig** in the left chamber, close the left solenoid valve by clicking the 'Left Solenoid' button in LabVIEW once. Note that **40 psig** means that we have a gage pressure of **40 psi**.
 - e. Using LabVIEW, decrease the Flow Rate to 0 g/min
 - f. Close Valve A2
4. Vacuum the right tank to **-6 psig**. The procedure to do this is as follows:
 - a. Open Valves A1 and A5.
 - b. Open the Right Solenoid valve by pressing the 'Right Solenoid' button in LabVIEW once.
 - c. Using LabVIEW, gradually increase the Flow Rate to 50 g/min.

- d. When the pressure reaches **-6 psig** in the right chamber, close the right solenoid valve by clicking the 'Right Solenoid' button in LabVIEW once. Note that **-6 psig** means that we have a gage pressure of **-6 psi**.
- e. Using LabVIEW, decrease the Flow Rate to 0 g/min.
- f. Close Valves A1 and A5.

Now you are going to let the tanks come to equilibrium in 2 different ways. The first way will involve using the center solenoid valve. The second way will involve using the micrometer needle valve.

Method 1 – Using the Center Solenoid Valve:

5. Use the LabVIEW program to open the center solenoid valve and leave it in the open position until there is an **insignificant temperature difference** between the two tanks (this should take roughly 45 seconds).
6. In LabVIEW, click 'Stop Recording' and save your data– labeling it **Lab 1 – Part 1a**
7. In LabVIEW, click 'Reset'.

Repeat the *Procedure* starting from Step 1 (on the previous page); this time using **Method 2** from step 5 and onwards. **REMEMBER TO EMPTY THE TANK FIRST** using the “Bar” valve behind the left tank (valves C1 and C2) and/or the black pressure-release valve (Valve B1).

Method 2 – Using the Micrometer Needle Valve:

5. Open the small ball valve (B2) and open the micrometer valve by turning in 4 times counterclockwise. Leave both valves open until there is an **insignificant pressure difference** between the two tanks (this should take roughly 10 minutes).
6. In LabVIEW, click 'Stop Recording' and save your data– labeling it **Lab 1 – Part 1b**
7. In LabVIEW, click 'Reset'.
8. Close both the small ball valve and the micrometer valve.

**** REMEMBER TO EMPTY THE TANK BEFORE MOVING ON TO PART 2 ****

Discussion (3 Questions)

1. Using the ideal gas law and conservation of mass determine the volume ratio between the two tanks. Hint: initial state conditions for Left and Right Tanks are P and T before expansion, final condition for the tanks is the equilibrium P and T after expansion.
2. Explain the heat transfer between the tanks and the surroundings for both of the expansion cases. What is the net heat transfer? Hint: Calculations are not required, only explain the process.
3. How does this experiment show that the state properties are path independent? Use a figure to assist your explanation.

Part 2: Determine the Initial Mass and Volume of the Left Tank

Introduction

In order to determine the initial mass, all we must do is determine how much mass has entered the tank. This is done by calculating the area underneath the mass flow rate graph. To determine the volume, we will use the ideal gas law, $PV=MRT$ and rearrange to solve for volume, $V=MRT/P$. Since all the variables are known (M, R, T, and P known) from the graphs we obtain, we can plug them into the formula and thus find the final volume.

To begin a lab:

1. Turn the computer on and start LabVIEW (or click 'Reset' button if LabVIEW is already running)
2. Make sure all valves are closed.

*Note: Recall from the previous lab that the **regulator pressure** will already be preset to 80 psi.

Procedure

1. Record the ambient pressure using the manometer in the lab.
2. Click the 'Start Collecting Data' button in LabVIEW to begin recording the results on the graphs.
3. Pressurize left tank to **40 Psig** and wait for P and T to stabilize. To do this:
 - a. Open Valve A2.
 - b. Open the Left Solenoid valve by pressing the 'Left Solenoid' button in LabVIEW once.
 - c. Using LabVIEW, gradually increase the Flow Rate to 50 g/min.
 - d. When the pressure reaches **40 psig** in the left chamber, click the 'Left Solenoid' button in LabVIEW once. Note that **40 psig** means that we have a gage pressure of **40 psi**.
 - e. Using LabVIEW, decrease the Flow Rate to 0 g/min
 - f. Close Valve A2
 - g. Wait for P and T to stabilize (wait for the values of pressure and temperature on the graphs in LabVIEW to remain constant with time)
4. In LabVIEW, click 'Stop Recording' and save your data— labeling it **Lab 1 – Part 2**
5. Empty the left tank using the bar valve and/or the black pressure-release valve (Valve B1).
6. In LabVIEW, click 'Reset'.

Discussion (3 Questions)

1. Calculate the initial mass and volume of the left tank.
2. What are the sources of error?
3. Does the compressibility of air affect your results?